



COMSATS University Islamabad Abbottabad Campus
Department of Computer Science

ASSIGNMENT 02

CLO-3 (CDF Unit # 3)

SOFTWARE PROJECT MANAGEMENT

Instructions

Please read the following instructions carefully before assignment submission.

- It should be clear that your **assignment will not get any credit if:**
 - The assignment is **submitted after due date.**
 - The submitted assignment is **not well formatted.**
 - The assignment is **plagiarized.**
 - The assignment submission via **email.**
 - The assignment should be **uploaded to the portal** and should be in word **document format.**
 - **Due date** of submission on portal is **31st May, 2026 at 11:59 pm.**
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Before attempting this assignment, read PMBOK: Chapter 8 carefully. Apply Chapter 8 concepts to the project you submitted in Assignment 1. Generic or AI-generated responses that could apply to any project will not be accepted. Draw diagrams using your own project details and label everything with terminology from your Assignment 1 document.

Questions:

1. Distinguish between quality and grade as defined in project quality management and explain how the Cost of Conformance and Cost of Non-Conformance each contribute to overall project cost. Give an example of prevention cost and external failure cost drawn from your selected software project context and summarize why prevention is preferred over inspection according to quality management principles.
2. Describe the contributions of Joseph Juran, W. Edwards Deming, and Philip Crosby to the evolution of quality management, and classify their key concepts — the 80/20 rule, the PDCA cycle, and Zero Defects — by associating each concept with the theorist who introduced it. Paraphrase how Deming's PDCA cycle supports the principle of continuous improvement (Kaizen) and predict what would happen to product quality in a project that skips the "Check" and "Act" stages of the cycle.
3. Draw a PDCA cycle diagram for your selected project showing how your software team would apply each stage to improve a repeated defect problem in their testing process. Label each stage in your diagram with at least three specific quality activities taken from your project, and summarize below the diagram how moving through the full cycle would reduce the repetition of that defect.
4. Interpret the difference between Quality Planning, Manage Quality, and Control Quality as three distinct process groups in project quality management. For each process, identify its primary output and demonstrate how one quality tool or technique used in that process contributes to meeting quality standards. Match each of the following tools — Pareto chart, cause-and-effect diagram, control chart etc. to the process group where it is most commonly applied and explain your reasoning for each choice.

5. Comprehend the difference between Quality Assurance and Quality Control by describing the focus, responsibility, and timing of each. Highlight from these differences why QA is considered a proactive process while QC is reactive and defends this conclusion with reference to at least one technique used in each process. Translate the concept of Total Quality Management (TQM) into your own words and generalize how its core principles generalize to agile project environments and give one example of how practices like retrospectives or small batch deliveries extend the TQM mindset in an agile team.
6. For a specific quality problem identified in your selected project draw a fishbone (cause-and-effect) diagram that maps out the possible root causes. Organize your diagram under the standard cause categories (as shown in lecture slides of chapter 08). Then, based on what your diagram shows whether the identified root causes are better addressed through Quality Assurance or Quality Control, and defend your choice with reference to at least three causes shown in your diagram.

Note:

- ✚ Do not paste copied text from books or online sources directly. Research and add content after thorough examination/study.
- ✚ Be precise and clear.
- ✚ Assignment should be in group. (4 members)
- ✚ Assignment should be attempted in sequence.
- ✚ The attempted assignment should not exceed more than 4 pages (excluding title page).

GOOD LUCK!!!!

The brain is the blessing of LORD, the more you use more it multiplies. 😊